

Digital Rock Physics (DRP) – Rockphys 2

UENF – Equinor

Bi-weekly meeting, 24-06-2026

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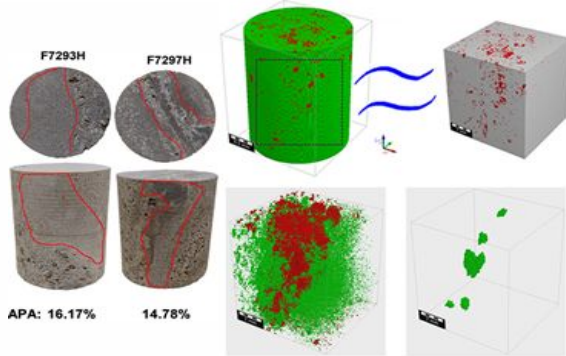


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Digital Rock Physics (DRP) – Rockphys2

- A brief review of the main activities:

1- Elastic Anisotropy Evaluation



Vertical Transverse Isotropic (VTI)

Horizontal Transverse Isotropic (HTI)

$\theta = 0^\circ$ (Z-axis):
 $V_s(0^\circ) = \text{Mean}(V_{sh}, V_{sv})$,
 $V_{sa} \approx V_{sb}$.

$V_p(0^\circ) \uparrow V_{sa}(0^\circ)$
 $V_{sb}(0^\circ) \downarrow$

$\theta = 90^\circ$:
 $V_p(90^\circ) = \text{Mean}(V_{px}, V_{py})$,
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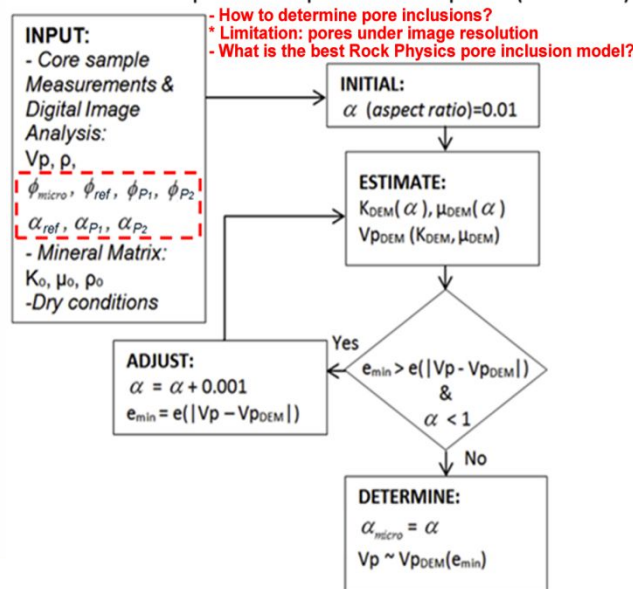
$V_p(90^\circ) \uparrow V_{sa}(90^\circ)$
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$\theta = 0^\circ$ (X or Y-axis):
 $V_p(0^\circ) < V_p(90^\circ)$,
 $\text{Min}(V_{px}, V_{py}) = V_p$,
 $V_{sa}, V_{sb}(0^\circ)$ at same axis, and
 $V_s(0^\circ) = \text{Mean}(V_{sa}, V_{sb})$,
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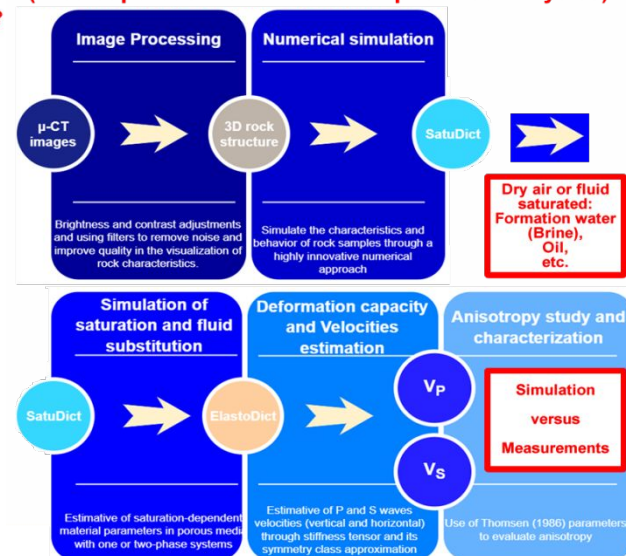
2- Rock Physics pore inclusion modeling

Multiscale: Multiple visible parts + micropores (invisible*)



3- Numerical Simulation for Fluid Substitution

Compare elastic parameters from dry and saturated Core Lab measures versus Digital Rock simulations (Predict porous media relationship for Rock Physics)



OUTLINE (Herson)

- Summarized Title

- I. Introduction
- II. Data set
- III. Methodology
- IV. Results
- V. Conclusions
- VI. Next steps



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I. Introduction (**Herson**)

- This study addresses a **comparative analysis** between variate of **elastic anisotropy** in samples **dry and saturated** using **Digital Rock Physics** model.
- Goals:
 - Analyze the **elastic behavior** of **carbonate rock samples** using **digital simulations** in **dry and saturated conditions**.
 - Extract and evaluate **stiffness tensors** through the **GeoDict software** (ElastoDict module).
 - Compute Thomsen **anisotropy parameters** (ϵ , δ , γ) and angular **wave velocities**.
 - Assess the **sensitivity of different symmetry** approximations (VTI and HTI).
 - Quantify **non-ellipticity parameters** deviations using η , η' , and η'' indicators.
 - Automate the post-processing workflow via Python scripting.



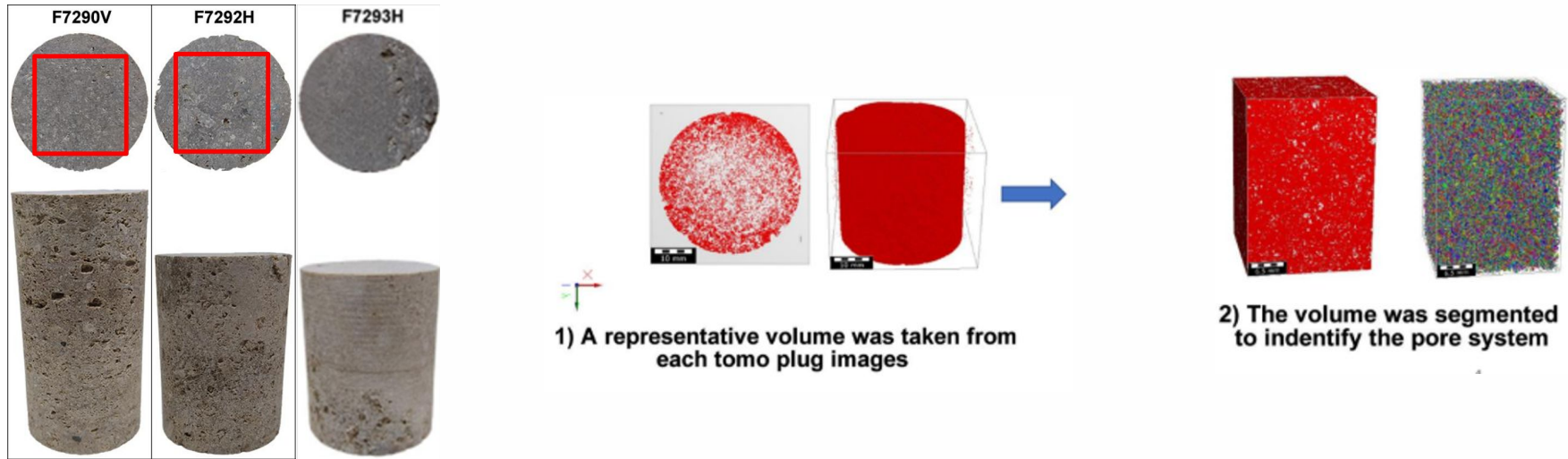
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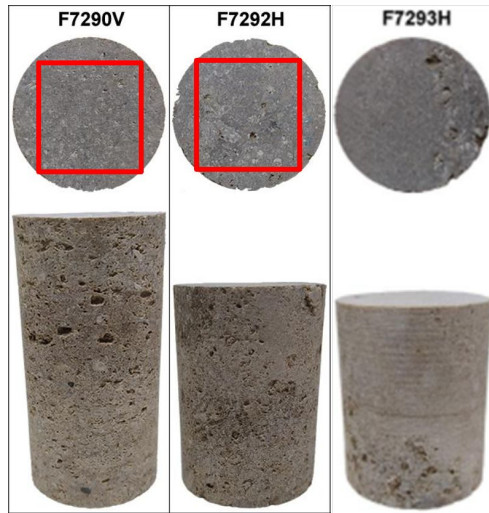
II. Data set (**Herson**)

- A total of 3 digital rock samples (**F7290V**, **F7292H**, and **F7293H**) obtained from Well #1045 (**1-BRSA-1045-SPS**) and processed via **μ CT imaging** for **numerical simulation**. Voxel resolution: **$\sim 12.14 \mu\text{m}$** , **1.5" inches in diameter**, and acquisition in PUC/RS. Simulation of velocities (**P and S waves**) of dry samples under a confining pressure of 22.1 MPa under similar conditions (**dry and sat.**) to those measured in the laboratory.



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III. Theory and method (Herson)

$$\begin{pmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{yz} \\ \sigma_{xz} \\ \sigma_{xy} \end{pmatrix} = \begin{pmatrix} C_{11} & C_{12} & C_{13} & C_{14} & C_{15} & C_{16} \\ C_{12} & C_{22} & C_{23} & C_{24} & C_{25} & C_{26} \\ C_{13} & C_{23} & C_{33} & C_{34} & C_{35} & C_{36} \\ C_{14} & C_{24} & C_{34} & C_{44} & C_{45} & C_{46} \\ C_{15} & C_{25} & C_{35} & C_{45} & C_{55} & C_{56} \\ C_{16} & C_{26} & C_{36} & C_{46} & C_{56} & C_{66} \end{pmatrix} \cdot \begin{pmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ 2\varepsilon_{yz} \\ 2\varepsilon_{xz} \\ 2\varepsilon_{xy} \end{pmatrix}$$

Stress **Elastic or Stiffness** **Strain**

Assuming the V_p and V_s for core plugs at $\theta = 0$ and 90° concerning the sample's axis symmetry, the elastic constants can be calculated:

$$C_{11} = \rho V_P(90^\circ)^2$$

$$C_{33} = \rho V_P(0^\circ)^2$$

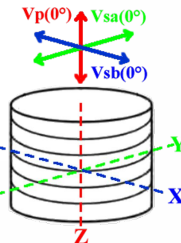
$$C_{44} = \rho V_S(0^\circ)^2 = \rho V_{Sv-slow}(90^\circ)^2$$

$$C_{66} = \rho V_{Sh-fast}(90^\circ)^2$$

Vertical Transverse Isotropic (VTI)

$\theta = 0^\circ$ (Z-axis):
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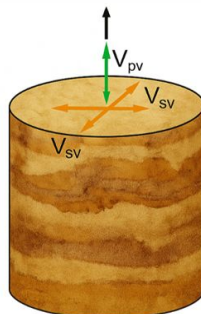
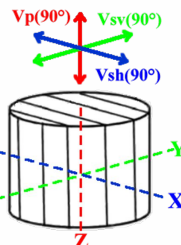
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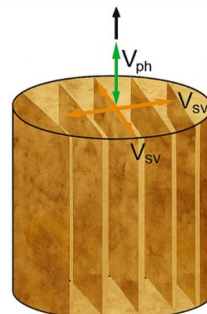
Horizontal Transverse Isotropic (HTI)

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$\theta = 0^\circ$ (X or Y-axis):
 $V_p(0^\circ) < V_p(90^\circ)$,
 $\text{Min}(V_{px}, V_{py}) = V_p, V_{sa}, V_{sb}(0^\circ)$
at same axis, and
 $V_s(0^\circ) = \text{Mean}(V_{sa}, V_{sb})$,
 $V_{sa} \approx V_{sb}$.



Vertical plug



Horizontal plug

Alkhalifah (1995)
Thomsen (1986)
Stovas & Ursin (2007)

$$v_{SH}(\theta) \approx v_{S0} \left[1 + \gamma \sin^2 \theta \right]$$

Wave velocities

- $V_{P0} = \sqrt{\frac{C_{33}}{\rho}}$
- $V_{S0} = \sqrt{\frac{C_{44}}{\rho}}$

Ellipticity indicators

$$\eta = \frac{\varepsilon - \delta}{1 + 2\delta} \quad (\text{Alkhalifah}) \quad \eta' = \varepsilon - \delta \quad (\text{Thomsen})$$

$$\eta'' = \frac{\varepsilon - \delta}{(1 + 2\delta)^2} \left(1 + \frac{2\gamma_0^2 \delta}{\gamma_0^2 - 1} \right) \quad (\text{Stovas and Ursin})$$

Angular velocity functions

$$v_p(\theta) \approx v_{p0} \left[1 + \delta \sin^2 \theta + \eta' \sin^4 \theta \right]$$

$$v_{sv}(\theta) \approx v_{s0} \left[1 + \left(\frac{v_{p0}}{v_{s0}} \right)^2 \eta' \sin^2 \theta + \left(\frac{v_{p0}}{v_{s0}} \right)^2 \eta' \sin^4 \theta \right]$$

III. Theory and method (**Herson**)



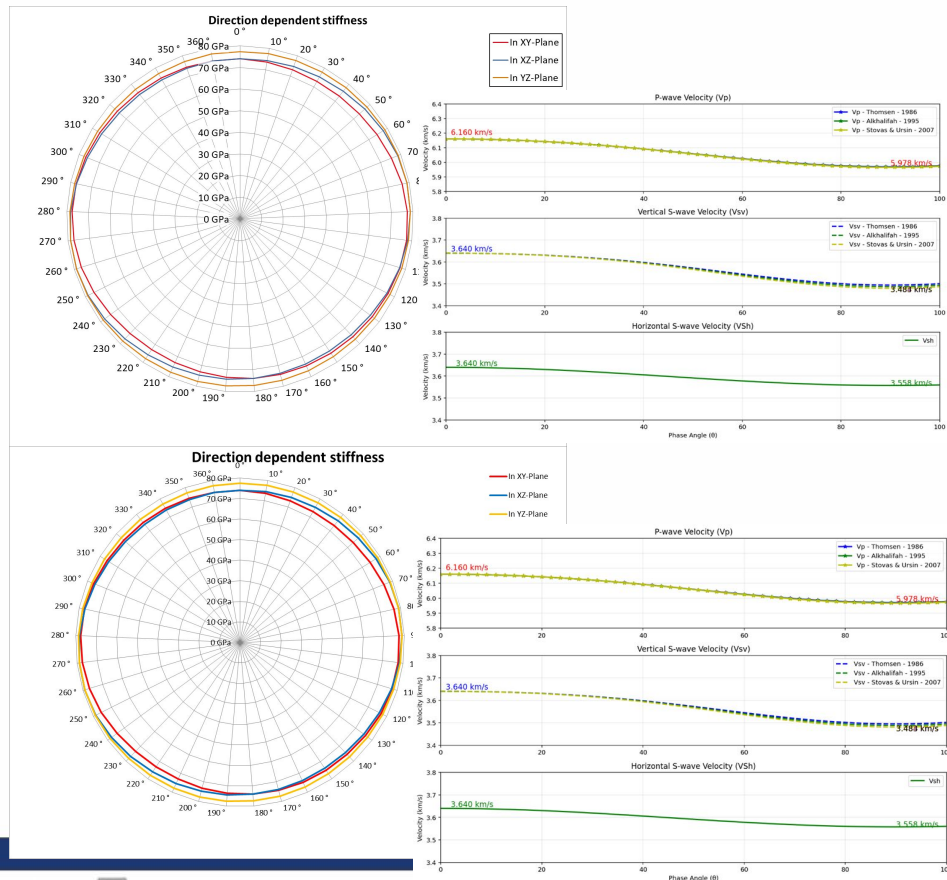
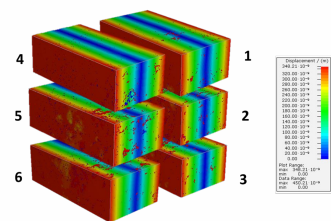
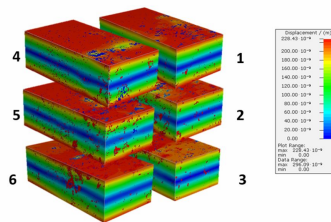
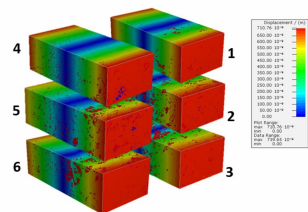
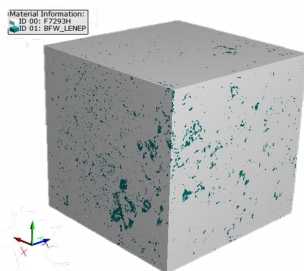
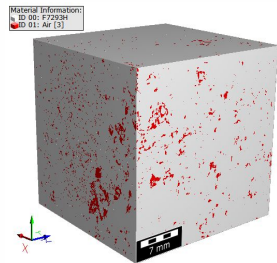
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IV. Results (Herson)

F7293H



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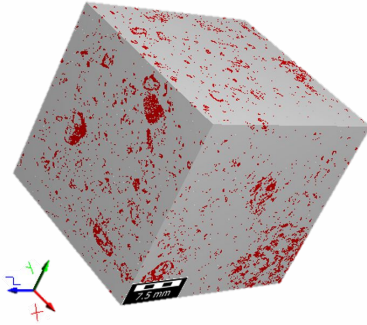


V. Conclusions (Herson)

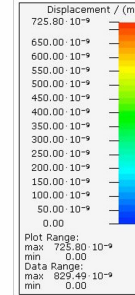
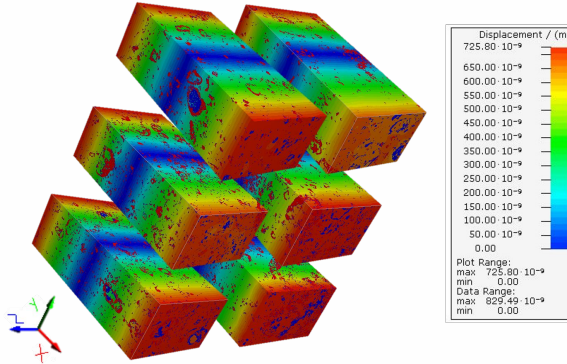
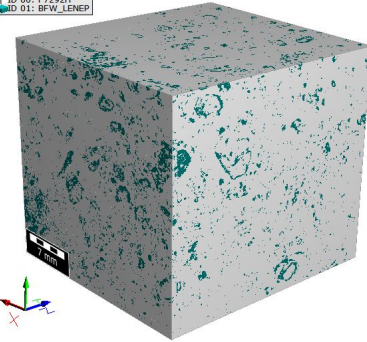
F7292H



Material Information:
ID 00: F7292H
ID 01: Air



Material Information:
ID 00: F7292H
ID 01: BFW_LENEP



VI. Next steps (**Herson**)



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I. Introduction (**Romulo**)

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 - Compute ...



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II. Data set (Romulo)

- A total of ...



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III. Theory and method (**Romulo**)



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IV. Results (Romulo)



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V. Conclusions (Romulo)



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VI. Next steps (Romulo)



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II. Data set (Irineu)

- A total of ...



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III. Theory and method (Irineu)



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IV. Results (Irineu)



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V. Conclusions (Irineu)



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VI. Next steps (Irineu)



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References

- All references



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Thank you!



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